

Project report

See terminal execution outputs in the cell output. The final output can be found at the bottom of the cell output.

```
In [1]: %run agentic_workflow.py
```

*** Workflow execution started ***

Task to complete in this workflow, workflow prompt = Define a project tasks for this product by creating user stories, key features, and development tasks step by step.

Defining workflow steps from the workflow prompt

Action planned by the planning agent

1. Generate user stories - create comprehensive user stories for the Email Router product
2. Define product features - create product features with Feature Name, Description, Key Functionality, and User Benefit
3. Create development tasks - create engineering tasks with Task ID, Title, Description, Acceptance Criteria, Estimated Effort, and Dependencies

step:

1. Generate user stories - create comprehensive user stories for the Email Router product

0.5189650003921271

0.5127425954842004

0.3569054576549945

[Router] Best agent: Product Manager (score=0.519)

--- Interaction 1 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

1. Generate user stories - create comprehensive user stories for the Email Router product

Previous output:

Calling OpenAI API-----

Model: gpt-3.5-turbo

Messages: [{'role': 'system', 'content': "\n\nYou are \nYou are a Product Manager.\nYou, as a product manager, are responsible for defining the user stories for a product.\n\nReturn 3-4 user stories in the form:\n1: As a [user type], I want [action/feature] so that [benefit/jobs to be done].\n2: As a [user type], I want [action/feature] so that [benefit/jobs to be done].\n3: As a [user type], I want [action/feature] so that [benefit/jobs to be done].\n(continue as many as you need)\n\nDo not include any functionality outside this product.\n\nSpecification:\n\n[PASTE EMAIL ROUTER SPEC HERE]\n\n\nAs a knowledge-based assistant, in your output, do not use your own knowledge, but use only the following knowledge : \nStories are defined by writing sentences with a persona, an action, and a desired outcome.\nThe sentences always start with: As a\nWrite several stories for the product spec based on your knowledge.\nThe personas are the different users of the product.\n\nYour knowledge is defined as below:\nProduct Specification Document\nEmail Router\n1. Introduction\n1.1 Business Overview\nOur organization handles a high volume of incoming email communications that require timely, accurate responses across multiple departments. Currently, manual email triage and routing creates bottlenecks, delays customer responses, and consumes valuable staff time that could be directed toward higher-value activities.\nTeam members spend significant portions of their workday sorting through emails, determining appropriate handlers, and forwarding messages—a process that introduces inconsistency in response quality and timing. Subject matter experts often receive misdirected inquiries, while routine questions that could be automated consume unnecessary attention.\n\nThe introduction of the Email Router system addresses these operational challenges."}]

enges by implementing AI-powered email analysis and routing. This solution will integrate with our existing email infrastructure to automatically categorize incoming messages, generate responses for routine inquiries, and intelligently route complex communications to the appropriate subject matter experts.

By implementing this system, we aim to reduce response times, ensure consistent communication quality, and free our team members to focus on specialized tasks requiring human expertise rather than email management.

1.2 Problem Statement

Our organization is struggling with inefficient email management processes that negatively impact operational performance and customer satisfaction. The current manual approach to email triage and routing presents several critical challenges:

- Significant time delays occur between receipt and response as emails pass through multiple hands before reaching the appropriate subject matter expert.
- Staff members across departments spend excessive time sorting, categorizing, and forwarding emails, diverting resources from high-value work requiring specialized expertise.
- Inconsistent handling of similar inquiries results in variable response quality and contradictory information being provided to customers and partners.
- Important emails may be overlooked in high-volume periods, leading to missed opportunities or unaddressed critical issues.
- The organization lacks visibility into email traffic patterns that could inform staffing decisions and process improvements.
- Routine, repetitive inquiries that could be automated consume disproportionate staff attention and resources.

Without an automated solution to streamline email classification and routing, these inefficiencies will continue to create operational bottlenecks, increase response times, and potentially damage our reputation for customer service excellence.

1.3 Current Process

The existing email management process in our organization follows a largely manual workflow that involves multiple touchpoints and handoffs:

- Incoming emails arrive at general departmental inboxes (e.g., info@company.com, support@company.com) or to specific team members.
- Administrative staff or designated team members manually open, read, and make initial assessments of each email to determine its priority and appropriate handler.
- For routine inquiries, staff members search for relevant information across various knowledge bases, documentation, or by consulting colleagues before drafting responses.
- More complex or specialized inquiries require forwarding to subject matter experts (SMEs), often involving multiple transfers before reaching the appropriate person.
- SMEs must review each email, including the chain of previous communications, to understand the context before responding.
- Responses are drafted individually, with limited standardization or templates, leading to inconsistent messaging and varying quality.
- Follow-up tracking is managed through ad-hoc methods such as flagging emails, creating calendar reminders, or maintaining separate spreadsheets.
- Performance metrics and response time data are collected manually, if at all, providing little insight into operational efficiency or areas for improvement.
- During peak periods or staff absences, backlogs develop quickly with no automated prioritization system to ensure critical communications receive prompt attention.
- Cross-departmental inquiries often bounce between teams with unclear ownership, further delaying resolution and creating customer frustration.

This labor-intensive process creates significant inefficiencies, delays customer communication, and represents a substantial opportunity cost as skilled staff members devote time to email management rather than applying their expertise to core business functions.

1.4 Scope

The Email Router project encompasses the development, deployment, and integration of an AI-powered email management system that will:

- Interface with our existing email infrastructure to capture and process all incoming external communications.
- Analyze email content using natural language processing to determine intent, urgency, and required expertise.
- Automatically generate responses for routine, standard inquiries based on approved organizational knowledge.
- Route complex inquiries to appropriate subject matter experts based on content analysis and defined business rules.
- Provide a management dashboard for monitoring system performance, workflow bottlenecks, and response metrics.
- Include tools for continuous improvement through feedback loops and model training.

The project scope specifically excludes:

- Modifications to existing backend email servers
- Processing of internal employee-to-employee communications
- Integration with CRM systems (planned for future phase)
- Mobile application development (web interface only in

initial release)\n\n1.5 Objectives\n\nPrimary Objectives\nReduce Response Time: Decrease average email response time by 60% within three months of full implementation.\nIncrease Efficiency: Automate responses to at least 40% of incoming emails that contain routine inquiries.\nImprove Routing Accuracy: Achieve 90% accuracy in routing emails to appropriate subject matter experts by the end of the pilot phase.\nEnhance Consistency: Standardize responses to common inquiries to ensure consistent messaging and information delivery.\nLiberate Staff Resources: Reduce time spent by SMEs and administrative staff on email triage by 70%, allowing reallocation to higher-value activities.\nSecondary Objectives\nGenerate Insights: Develop analytics capabilities to identify communication trends, common customer pain points, and potential service improvements.\nIncrease Scalability: Create a solution that can accommodate 200% growth in email volume without proportional increases in staffing.\nImprove Customer Satisfaction: Achieve a 30% improvement in customer satisfaction metrics related to communication responsiveness.\nSupport Knowledge Management: Identify gaps in current knowledge base through analysis of inquiries that cannot be automatically answered.\nEnhance Compliance: Ensure all communications adhere to organizational standards and regulatory requirements through consistent handling.\n\n2. Product Overview\n2.1 Product Features\nEmail Ingestion System\nSeamless integration with email services via SMTP, IMAP, and RESTful APIs.\nReal-time email retrieval and preprocessing to extract relevant metadata and content.\nMessage Classification Module\nUtilization of LLM-based classifiers to analyze email content and determine intent and category.\nAssignment of confidence scores to decide between automated responses and manual handling.\nKnowledge Base Integration\nImplementation of a vector database for efficient storage and retrieval of organizational knowledge.\nContinuous learning mechanism to update the knowledge base with new information from resolved inquiries.\nResponse Generation Engine\nDeployment of a RAG system to generate contextually accurate and human-like responses.\nIncorporation of an approval workflow for reviewing and editing automated responses before dispatch.\nRouting Logic\nDevelopment of a rules-based engine to assign emails to appropriate SMEs based on content analysis.\nContext-aware forwarding that includes relevant metadata and previous correspondence history.\nUser Interface\nCreation of a comprehensive dashboard for monitoring system performance, including metrics on response times and accuracy.\nProvision of a configuration panel for managing the knowledge base, routing rules, and system settings.\nImplementation of manual override options to allow human intervention when necessary.\n2.2 User Classes and Characteristics\nCustomer Support Representatives: Will benefit from reduced workload on routine inquiries, allowing focus on complex issues.\nSubject Matter Experts (SMEs): Will receive only relevant, complex inquiries, improving efficiency and job satisfaction.\nIT Administrators: Responsible for system configuration, maintenance, and monitoring performance metrics.\n2.3 Operating Environment\nThe Email Router will operate within the organization's existing IT infrastructure, integrating with standard email services and data bases. It will be deployed on secure servers, adhering to the organization's data protection and privacy policies.\n2.4 Constraints\nData Privacy: Compliance with data protection regulations is mandatory, ensuring that sensitive information is handled appropriately.\nIntegration: The system must seamlessly integrate with existing email platforms without requiring significant changes to current workflows.\nScalability: The architecture should support scaling to accommodate increasing email volumes as the organization grows.\n3. Functional Requirements\nEmail Ingestion\nThe system shall connect to designated email services to retrieve incoming messages in real-time.\nThe system shall preprocess emails to extract metadata such as sender, recipient, subject, and timestamp.\nMessage Classification\nThe system shall analyze email content to determine intent and category using LLM-based models.\nThe system shall assign a confidence score to each classification to guide subsequent actions.\nKnowledge Base Retrieval\nThe system shall search the knowledge base for relevant information corresponding to the classified intent.\nThe system shall update the knowledge base with new information from resolved inquiries.\nResponse Generation\nThe system shall generate draft responses for routine inquiries using the RAG system.\nThe system shall provide an interface for human review and approval of generated responses.\nEmail Routing\nThe system shall fo


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step:
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2. Define product features - create product features with Feature Name, Description, Key Functionality, and User Benefit

0.40117391107189543

0.39904753567473117

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--- Interaction 1 ---

```

Prompt:

```
Previous output: 1: As a Customer Support Representative, I want the Email Router
system to automatically categorize incoming messages so that I can focus on handl
ing complex customer inquiries efficiently.
```

3: As an IT Administrator, I want the Email Router system to provide a dashboard displaying performance metrics so that I can monitor system performance and make necessary adjustments for optimal operation.

Calling OpenAI API-----

```

Messages: [{'role': 'system', 'content': '\n
You are \nYou are a Program
Manager, you are responsible for defining feature from user stories.\n\n
As a knowledge-based assistant, in your output, do not use your own knowledge, bu
t use only the following knowledge : \nYour output consists only of list of featu
res.\nFeatures are created from user stories step by step as below:\nstep 1. Find
a user story statement from the Previous output.\nstep 2. Find a type of user aft
er "As a" statement in the user story.\nstep 3. Find an action or feature after
"I want" statement in the user story..\nstep 4. Find benefit/value after "so tha
t" statement in the user story..\nstep 5. iterate step 1 to step 4 until all user
stories are picked up.\nstep 6. Create features in structure below from the user
stories.\n- Feature Name: [Insert a clear, concise title that identifies the capa
bility]\n- Description: [Insert a brief explanation of what the feature does and
its purpose]\n- Key Functionality: [Insert the specific capabilities or actions t
he feature provides]\n- User Benefit: [Insert how this feature creates value for
the user]\n- User Story: [Insert a user story that describes how the feature will
be used]\n\nThe user stories are given in the user prompt after a line which incl
udes "Previous ouput:". \n1: As a [type of user], I want [an action or feature] so
that [benefit/value]\n2: As a [type of user], I want [an action or feature] so th
at [benefit/value]\n3: As a [type of user], I want [an action or feature] so that
[benefit/value]\n(continues)\n\n
\n\n Forget previous context.'}, {'role':
'user', 'content': 'User stories are defined in 2. Define product features - crea
te product features with Feature Name, Description, Key Functionality, and User B
enefit\n\nPrevious output: 1: As a Customer Support Representative, I want the Em
ail Router system to automatically categorize incoming messages so that I can foc
us on handling complex customer inquiries efficiently.\n2: As a Subject Matter Ex
pert (SME), I want the Email Router system to intelligently route complex communi

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cations to me based on content analysis so that I can address inquiries that require my expertise promptly.\n3: As an IT Administrator, I want the Email Router system to provide a dashboard displaying performance metrics so that I can monitor system performance and make necessary adjustments for optimal operation.\n4: As a Team Leader, I want the Email Router system to standardize responses to common inquiries to ensure consistent messaging and information delivery across all customer interactions.'}}]

Worker Agent Response:

- Feature Name: Automatic Categorization of Incoming Messages
- Description: This feature automatically categorizes incoming messages within the Email Router system.
- Key Functionality: The system analyzes the content of incoming messages and assigns appropriate categories to them.
- User Benefit: Customer Support Representatives can efficiently handle complex customer inquiries by focusing on messages that are already categorized.

- Feature Name: Intelligent Routing of Complex Communications
- Description: This feature intelligently routes complex communications to Subject Matter Experts based on content analysis.
- Key Functionality: The system identifies messages that require expertise and directs them to the relevant SME for prompt attention.
- User Benefit: Subject Matter Experts can address inquiries that specifically require their expertise promptly, improving overall response time and customer satisfaction.

- Feature Name: Performance Metrics Dashboard
- Description: This feature provides a dashboard within the Email Router system displaying performance metrics.
- Key Functionality: The dashboard shows key performance indicators and system metrics for monitoring and analysis.
- User Benefit: IT Administrators can monitor system performance, identify areas for improvement, and make necessary adjustments for optimal operation.

- Feature Name: Standardized Responses for Common Inquiries
- Description: This feature enables the Email Router system to standardize responses to common inquiries.
- Key Functionality: The system provides predefined responses for common inquiries to ensure consistent messaging across all customer interactions.
- User Benefit: Team Leaders can maintain consistent messaging and information delivery across all customer interactions, enhancing the overall customer experience.

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

Yes, the answer meets the criteria. Each feature is clearly structured with a Feature Name, Description, Key Functionality, User Benefit, and User Story, providing a comprehensive overview of the capabilities and benefits of each feature.

Step 3: Check if evaluation is positive

☒ Final solution accepted.

result:

- Feature Name: Automatic Categorization of Incoming Messages
- Description: This feature automatically categorizes incoming messages within the Email Router system.
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- User Benefit: Customer Support Representatives can efficiently handle complex c

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- User Benefit: IT Administrators can monitor system performance, identify areas for improvement, and make necessary adjustments for optimal operation.

- Feature Name: Standardized Responses for Common Inquiries
- Description: This feature enables the Email Router system to standardize responses to common inquiries.
- Key Functionality: The system provides predefined responses for common inquiries to ensure consistent messaging across all customer interactions.
- User Benefit: Team Leaders can maintain consistent messaging and information delivery across all customer interactions, enhancing the overall customer experience.

step:

3. Create development tasks - create engineering tasks with Task ID, Title, Description, Acceptance Criteria, Estimated Effort, and Dependencies

0.3463613548567957

0.3539672843431055

0.5556714478827375

[Router] Best agent: Development Engineer (score=0.556)

--- Interaction 1 ---

Step 1: Worker agent generates a response to the prompt

Prompt:

Features are defined in 3. Create development tasks - create engineering tasks with Task ID, Title, Description, Acceptance Criteria, Estimated Effort, and Dependencies

Previous output: - Feature Name: Automatic Categorization of Incoming Messages

- Description: This feature automatically categorizes incoming messages within the Email Router system.
- Key Functionality: The system analyzes the content of incoming messages and assigns appropriate categories to them.
- User Benefit: Customer Support Representatives can efficiently handle complex customer inquiries by focusing on messages that are already categorized.

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- Description: This feature intelligently routes complex communications to Subject Matter Experts based on content analysis.
- Key Functionality: The system identifies messages that require expertise and di

s feature intelligently routes complex communications to Subject Matter Experts based on content analysis.\n- Key Functionality: The system identifies messages that require expertise and directs them to the relevant SME for prompt attention.\n- User Benefit: Subject Matter Experts can address inquiries that specifically require their expertise promptly, improving overall response time and customer satisfaction.\n\n- Feature Name: Performance Metrics Dashboard\n- Description: This feature provides a dashboard within the Email Router system displaying performance metrics.\n- Key Functionality: The dashboard shows key performance indicators and system metrics for monitoring and analysis.\n- User Benefit: IT Administrators can monitor system performance, identify areas for improvement, and make necessary adjustments for optimal operation.\n\n- Feature Name: Standardized Responses for Common Inquiries\n- Description: This feature enables the Email Router system to standardize responses to common inquiries.\n- Key Functionality: The system provides predefined responses for common inquiries to ensure consistent messaging across all customer interactions.\n- User Benefit: Team Leaders can maintain consistent messaging and information delivery across all customer interactions, enhancing the overall customer experience.'}]

Worker Agent Response:
Development Tasks:

1. Task ID: AUTOCAT-01

- Task Title: Implement Message Content Analysis
- Related User Story: As a Customer Support Representative, I want incoming messages to be automatically categorized so that I can efficiently handle complex customer inquiries.
- Description: Develop the functionality to analyze the content of incoming messages and assign appropriate categories to them within the Email Router system.
- Acceptance Criteria: Incoming messages are accurately categorized based on their content. The categorization process is automated and does not require manual intervention.
- Estimated Effort: 5 days
- Dependencies: None

2. Task ID: AUTOCAT-02

- Task Title: Configure Category Assignment Logic
- Related User Story: As a Customer Support Representative, I want the system to assign appropriate categories to incoming messages so that I can focus on categorized messages.
- Description: Define and implement the logic for assigning categories to incoming messages based on content analysis results.
- Acceptance Criteria: Messages are assigned categories according to predefined criteria. The logic is configurable and can be adjusted as needed.
- Estimated Effort: 3 days
- Dependencies: AUTOCAT-01

3. Task ID: INTRT-01

- Task Title: Develop Expertise Identification Algorithm
- Related User Story: As a Subject Matter Expert, I want complex communications to be routed to me based on content analysis so that I can address inquiries promptly.
- Description: Create an algorithm to identify messages that require expertise and route them to the relevant Subject Matter Experts for prompt attention.
- Acceptance Criteria: Messages requiring expertise are accurately identified and routed to the appropriate SMEs. The routing process is automated and efficient.
- Estimated Effort: 6 days
- Dependencies: None

4. Task ID: INTRT-02

- Task Title: Implement Routing Rules Configuration
- Related User Story: As a Subject Matter Expert, I want the system to direct inquiries requiring my expertise to me for prompt attention.
- Description: Develop the functionality to configure routing rules for directing messages to the relevant SMEs based on expertise requirements.
- Acceptance Criteria: Routing rules are defined and applied correctly to route messages to the appropriate SMEs. The configuration interface is user-friendly.
- Estimated Effort: 4 days
- Dependencies: INTRT-01

5. Task ID: PMD-01

- Task Title: Design Performance Metrics Dashboard UI
- Related User Story: As an IT Administrator, I want a dashboard displaying system performance metrics so that I can monitor and analyze system operation.
- Description: Design the user interface for the Performance Metrics Dashboard within the Email Router system.
- Acceptance Criteria: The dashboard UI includes key performance indicators and system metrics for monitoring. The design is intuitive and visually appealing.
- Estimated Effort: 3 days
- Dependencies: None

6. Task ID: PMD-02

- Task Title: Implement Data Visualization for Performance Metrics
- Related User Story: As an IT Administrator, I want to monitor system performance through visual representations on the dashboard.
- Description: Implement data visualization components to display performance metrics and system data on the Performance Metrics Dashboard.
- Acceptance Criteria: Performance metrics are presented graphically for easy interpretation. The data visualization is interactive and allows for detailed analysis.
- Estimated Effort: 5 days
- Dependencies: PMD-01

7. Task ID: STDRSP-01

- Task Title: Define Predefined Responses Structure
- Related User Story: As a Team Leader, I want standardized responses for common inquiries to ensure consistent messaging across customer interactions.
- Description: Define the structure and format for storing predefined responses to common inquiries within the Email Router system.
- Acceptance Criteria: Predefined responses are structured in a way that allows easy retrieval and utilization. The format supports customization and updates.
- Estimated Effort: 2 days
- Dependencies: None

8. Task ID: STDRSP-02

- Task Title: Implement Response Retrieval Mechanism
- Related User Story: As a Team Leader, I want the system to provide predefined responses for common inquiries to maintain consistent messaging.
- Description: Develop the mechanism to retrieve predefined responses and suggest them for common inquiries during customer interactions.
- Acceptance Criteria: The system accurately retrieves predefined responses based on inquiry context. Suggestions are relevant and enhance response consistency.
- Estimated Effort: 4 days
- Dependencies: STDRSP-01

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

Yes, the provided answer meets the criteria. It includes a collection of tasks, each defined with a unique Task ID, Task Title, Related User Story, Description, Acceptance Criteria, Estimated Effort, and Dependencies. The tasks are structured

in a clear and organized manner, addressing different user stories and requirements within the Email Router system.

Step 3: Check if evaluation is positive

☑ Final solution accepted.

result:

Development Tasks:

1. Task ID: AUTOCAT-01

- Task Title: Implement Message Content Analysis
- Related User Story: As a Customer Support Representative, I want incoming messages to be automatically categorized so that I can efficiently handle complex customer inquiries.
- Description: Develop the functionality to analyze the content of incoming messages and assign appropriate categories to them within the Email Router system.
- Acceptance Criteria: Incoming messages are accurately categorized based on their content. The categorization process is automated and does not require manual intervention.
- Estimated Effort: 5 days
- Dependencies: None

2. Task ID: AUTOCAT-02

- Task Title: Configure Category Assignment Logic
- Related User Story: As a Customer Support Representative, I want the system to assign appropriate categories to incoming messages so that I can focus on categorized messages.
- Description: Define and implement the logic for assigning categories to incoming messages based on content analysis results.
- Acceptance Criteria: Messages are assigned categories according to predefined criteria. The logic is configurable and can be adjusted as needed.
- Estimated Effort: 3 days
- Dependencies: AUTOCAT-01

3. Task ID: INTRT-01

- Task Title: Develop Expertise Identification Algorithm
- Related User Story: As a Subject Matter Expert, I want complex communications to be routed to me based on content analysis so that I can address inquiries promptly.
- Description: Create an algorithm to identify messages that require expertise and route them to the relevant Subject Matter Experts for prompt attention.
- Acceptance Criteria: Messages requiring expertise are accurately identified and routed to the appropriate SMEs. The routing process is automated and efficient.
- Estimated Effort: 6 days
- Dependencies: None

4. Task ID: INTRT-02

- Task Title: Implement Routing Rules Configuration
- Related User Story: As a Subject Matter Expert, I want the system to direct inquiries requiring my expertise to me for prompt attention.
- Description: Develop the functionality to configure routing rules for directing messages to the relevant SMEs based on expertise requirements.
- Acceptance Criteria: Routing rules are defined and applied correctly to route messages to the appropriate SMEs. The configuration interface is user-friendly.
- Estimated Effort: 4 days
- Dependencies: INTRT-01

5. Task ID: PMD-01

- Task Title: Design Performance Metrics Dashboard UI
- Related User Story: As an IT Administrator, I want a dashboard displaying system

m performance metrics so that I can monitor and analyze system operation.

- Description: Design the user interface for the Performance Metrics Dashboard within the Email Router system.
- Acceptance Criteria: The dashboard UI includes key performance indicators and system metrics for monitoring. The design is intuitive and visually appealing.
- Estimated Effort: 3 days
- Dependencies: None

6. Task ID: PMD-02

- Task Title: Implement Data Visualization for Performance Metrics
- Related User Story: As an IT Administrator, I want to monitor system performance through visual representations on the dashboard.
- Description: Implement data visualization components to display performance metrics and system data on the Performance Metrics Dashboard.
- Acceptance Criteria: Performance metrics are presented graphically for easy interpretation. The data visualization is interactive and allows for detailed analysis.
- Estimated Effort: 5 days
- Dependencies: PMD-01

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- Task Title: Define Predefined Responses Structure
- Related User Story: As a Team Leader, I want standardized responses for common inquiries to ensure consistent messaging across customer interactions.
- Description: Define the structure and format for storing predefined responses to common inquiries within the Email Router system.
- Acceptance Criteria: Predefined responses are structured in a way that allows easy retrieval and utilization. The format supports customization and updates.
- Estimated Effort: 2 days
- Dependencies: None

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- Related User Story: As a Team Leader, I want the system to provide predefined responses for common inquiries to maintain consistent messaging.
- Description: Develop the mechanism to retrieve predefined responses and suggest them for common inquiries during customer interactions.
- Acceptance Criteria: The system accurately retrieves predefined responses based on inquiry context. Suggestions are relevant and enhance response consistency.
- Estimated Effort: 4 days
- Dependencies: STDRSP-01

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final output

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step: 1. Generate user stories - create comprehensive user stories for the Email Router product

1: As a Customer Support Representative, I want the Email Router system to automatically categorize incoming messages so that I can focus on handling complex customer inquiries efficiently.

2: As a Subject Matter Expert (SME), I want the Email Router system to intelligently route complex communications to me based on content analysis so that I can address inquiries that require my expertise promptly.

3: As an IT Administrator, I want the Email Router system to provide a dashboard displaying performance metrics so that I can monitor system performance and make

necessary adjustments for optimal operation.

4: As a Team Leader, I want the Email Router system to standardize responses to common inquiries to ensure consistent messaging and information delivery across all customer interactions.

step: 2. Define product features - create product features with Feature Name, Description, Key Functionality, and User Benefit

- Feature Name: Automatic Categorization of Incoming Messages
- Description: This feature automatically categorizes incoming messages within the Email Router system.
- Key Functionality: The system analyzes the content of incoming messages and assigns appropriate categories to them.
- User Benefit: Customer Support Representatives can efficiently handle complex customer inquiries by focusing on messages that are already categorized.

- Feature Name: Intelligent Routing of Complex Communications
- Description: This feature intelligently routes complex communications to Subject Matter Experts based on content analysis.
- Key Functionality: The system identifies messages that require expertise and directs them to the relevant SME for prompt attention.
- User Benefit: Subject Matter Experts can address inquiries that specifically require their expertise promptly, improving overall response time and customer satisfaction.

- Feature Name: Performance Metrics Dashboard
- Description: This feature provides a dashboard within the Email Router system displaying performance metrics.
- Key Functionality: The dashboard shows key performance indicators and system metrics for monitoring and analysis.
- User Benefit: IT Administrators can monitor system performance, identify areas for improvement, and make necessary adjustments for optimal operation.

- Feature Name: Standardized Responses for Common Inquiries
- Description: This feature enables the Email Router system to standardize responses to common inquiries.
- Key Functionality: The system provides predefined responses for common inquiries to ensure consistent messaging across all customer interactions.
- User Benefit: Team Leaders can maintain consistent messaging and information delivery across all customer interactions, enhancing the overall customer experience.

step: 3. Create development tasks - create engineering tasks with Task ID, Title, Description, Acceptance Criteria, Estimated Effort, and Dependencies

Development Tasks:

1. Task ID: AUTOCAT-01

- Task Title: Implement Message Content Analysis
- Related User Story: As a Customer Support Representative, I want incoming messages to be automatically categorized so that I can efficiently handle complex customer inquiries.
- Description: Develop the functionality to analyze the content of incoming messages and assign appropriate categories to them within the Email Router system.

- Acceptance Criteria: Incoming messages are accurately categorized based on their content. The categorization process is automated and does not require manual intervention.
- Estimated Effort: 5 days
- Dependencies: None

2. Task ID: AUTOCAT-02

- Task Title: Configure Category Assignment Logic
- Related User Story: As a Customer Support Representative, I want the system to assign appropriate categories to incoming messages so that I can focus on categorized messages.
- Description: Define and implement the logic for assigning categories to incoming messages based on content analysis results.
- Acceptance Criteria: Messages are assigned categories according to predefined criteria. The logic is configurable and can be adjusted as needed.
- Estimated Effort: 3 days
- Dependencies: AUTOCAT-01

3. Task ID: INTRT-01

- Task Title: Develop Expertise Identification Algorithm
- Related User Story: As a Subject Matter Expert, I want complex communications to be routed to me based on content analysis so that I can address inquiries promptly.
- Description: Create an algorithm to identify messages that require expertise and route them to the relevant Subject Matter Experts for prompt attention.
- Acceptance Criteria: Messages requiring expertise are accurately identified and routed to the appropriate SMEs. The routing process is automated and efficient.
- Estimated Effort: 6 days
- Dependencies: None

4. Task ID: INTRT-02

- Task Title: Implement Routing Rules Configuration
- Related User Story: As a Subject Matter Expert, I want the system to direct inquiries requiring my expertise to me for prompt attention.
- Description: Develop the functionality to configure routing rules for directing messages to the relevant SMEs based on expertise requirements.
- Acceptance Criteria: Routing rules are defined and applied correctly to route messages to the appropriate SMEs. The configuration interface is user-friendly.
- Estimated Effort: 4 days
- Dependencies: INTRT-01

5. Task ID: PMD-01

- Task Title: Design Performance Metrics Dashboard UI
- Related User Story: As an IT Administrator, I want a dashboard displaying system performance metrics so that I can monitor and analyze system operation.
- Description: Design the user interface for the Performance Metrics Dashboard within the Email Router system.
- Acceptance Criteria: The dashboard UI includes key performance indicators and system metrics for monitoring. The design is intuitive and visually appealing.
- Estimated Effort: 3 days
- Dependencies: None

6. Task ID: PMD-02

- Task Title: Implement Data Visualization for Performance Metrics
- Related User Story: As an IT Administrator, I want to monitor system performance through visual representations on the dashboard.
- Description: Implement data visualization components to display performance metrics and system data on the Performance Metrics Dashboard.
- Acceptance Criteria: Performance metrics are presented graphically for easy interpretation. The data visualization is interactive and allows for detailed analysis.

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- Estimated Effort: 5 days
- Dependencies: PMD-01

7. Task ID: STDRSP-01

- Task Title: Define Predefined Responses Structure
- Related User Story: As a Team Leader, I want standardized responses for common inquiries to ensure consistent messaging across customer interactions.
- Description: Define the structure and format for storing predefined responses to common inquiries within the Email Router system.
- Acceptance Criteria: Predefined responses are structured in a way that allows easy retrieval and utilization. The format supports customization and updates.
- Estimated Effort: 2 days
- Dependencies: None

8. Task ID: STDRSP-02

- Task Title: Implement Response Retrieval Mechanism
- Related User Story: As a Team Leader, I want the system to provide predefined responses for common inquiries to maintain consistent messaging.
- Description: Develop the mechanism to retrieve predefined responses and suggest them for common inquiries during customer interactions.
- Acceptance Criteria: The system accurately retrieves predefined responses based on inquiry context. Suggestions are relevant and enhance response consistency.
- Estimated Effort: 4 days
- Dependencies: STDRSP-01

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